Observed Data Only

Incomplete Table

A	B	C	D
1	0.5	?	2.1
0	1.2	3.3	?
1	0.7	2.8	1.9

Missingness Mask

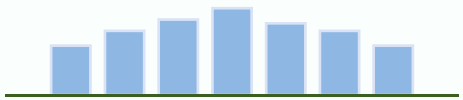
A	B	C	D
0	0	1	0
0	0	0	1
0	0	0	0

1=missing

Cross-Validation Prediction

Other columns → Target Column

Observed Residuals



$$r_j = x_j - \hat{f}(x_{-j})$$

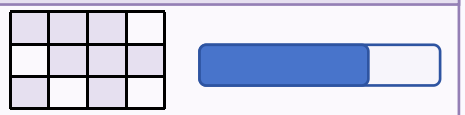
Observed Target Entries Only

No Access to Missing Values

MissDx: Missingness Diagnosis & Its Limits

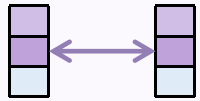
1.Diagnostic Evidence

Rate



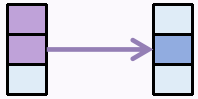
Per-Column Missingness

Mask-Mask



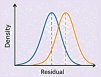
Dependence Between Masks

Mask-Value



Association with Observed Covariates

Conditional Residual



Shift

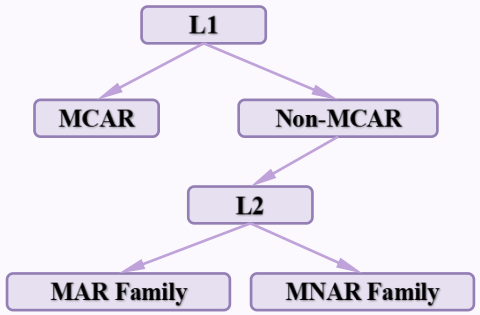


Skewness



kurtosis

2.Level-Structured Diagnosis



Diagnostic Evidence ≠ Identification

Insufficient Evidence

Sensitivity Analysis

MAR Imputation+Pertubation
Re-estimate Downstream Results

External References may provide additional evidence

3. Limits & Fallback

- Heterogeneous Marginals
- Weak Cross-column Predictability



Weaker Residual-shape Siganal

- Distribution Shift

MechZoo Evaluation

9 mechanism families · Synthetic + real-source tables with injected missingness

In-dist	Best-fresh	Held-out	Real-inj	Joint-shift
900	600	300	432	432

Labels reflect the mechanism actually applied. Structural-skip tables without the required driver column are labeled MCAR.